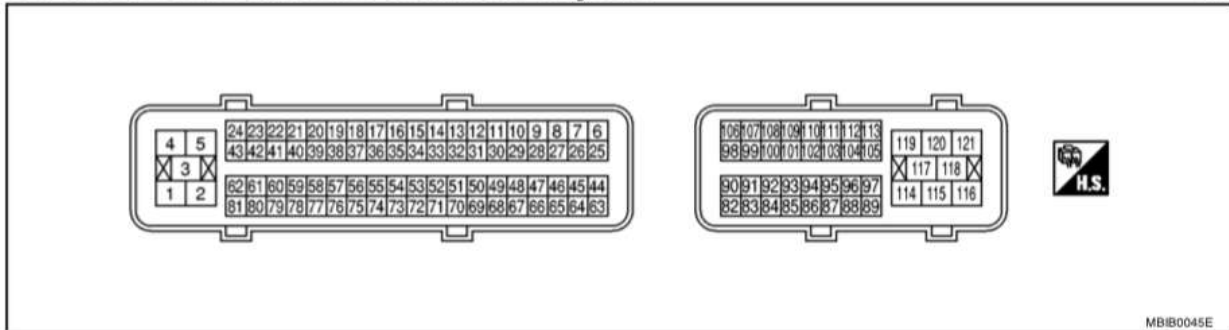


TROUBLE DIAGNOSIS

[CR (WITH EURO-OBD)]

ECM Harness Connector Terminal Layout

EBS0003N

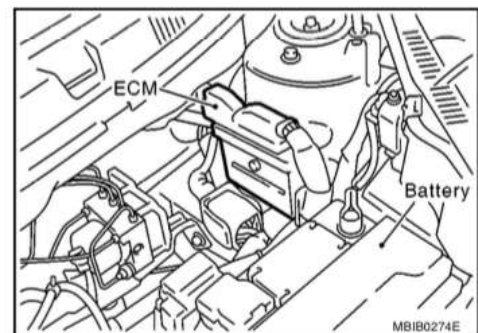


MBIB0045E

ECM Terminals and Reference Value PREPARATION

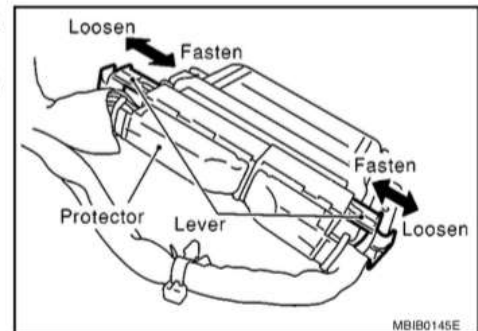
EBS0003O

1. ECM is located left hand side of the engine room.
2. Remove ECM harness protector.



MBIB0274E

3. When disconnecting ECM harness connector, loosen it with levers as far as they will go as shown at right.
4. Connect a break-out box (SST) and Y-cable adapter (SST) between the ECM and ECM harness connector.
 - Use extreme care not to touch 2 pins at one time.
 - Data is for comparison and may not be exact.



MBIB0145E

ECM INSPECTION TABLE

Specification data are reference values and are measured between each terminal and ground.
Pulse signal is measured by CONSULT-II.

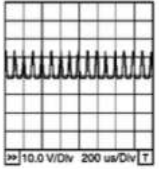
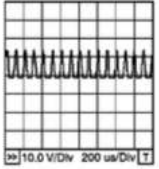
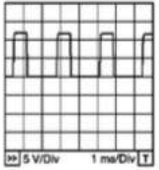
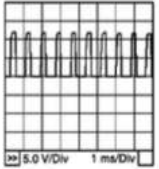
CAUTION:

Do not use ECM ground terminals when measuring input/output voltage. Doing so may result in damage to the ECM's transistor. Use a ground other than ECM terminals, such as the ground.

TERMI- NAL NO.	WIRE COLOR	ITEM	CONDITION	DATA (DC Voltage)
1	B	ECM ground	[Engine is running] • Idle speed	Engine ground

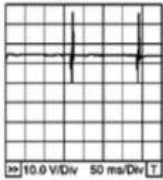
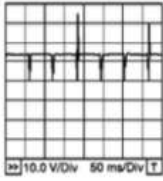
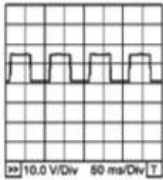
TROUBLE DIAGNOSIS

[CR (WITH EURO-OBD)]

TERMI- NAL NO.	WIRE COLOR	ITEM	CONDITION	DATA (DC Voltage)	
2	GY	Heated oxygen sensor 2 heater	[Engine is running] - Engine speed is below 3,600 rpm (A/T models), 3,800 rpm (M/T models) after the following conditions are met. - Engine: After warming up - Keeping the engine speed between 3,500 and 4,000 rpm for 1 minute and at idle for 1 minute under no load.	0 - 1.0V	A
			[Ignition switch ON] - Engine stopped [Engine is running] - Engine speed is above 3,600 rpm (A/T models), 3,800 rpm (M/T models)	BATTERY VOLTAGE (11 - 14V)	EC
3	LG	Throttle control motor relay power supply	[Ignition switch ON]	BATTERY VOLTAGE (11 - 14V)	C
4	L	Throttle control motor (Close)	[Ignition switch ON] Engine stopped - Shift lever position: D (A/T models) - Shift lever position: 1st (M/T models) - Accelerator pedal: Released	0 - 14V★  PBIB0534E	D
5	P	Throttle control motor (Open)	[Ignition switch ON] Engine stopped - Shift lever position: D (A/T models) - Shift lever position: 1st (M/T models) - Accelerator pedal: Fully depressed	0 - 14V★  PBIB0533E	E
13	Y	Crankshaft position sensor (POS)	[Engine is running] Warm-up condition - Idle speed NOTE: The pulse cycle changes depending on rpm at idle.	Approximately 3.0V★  PBIB0527E	F
			[Engine is running] Engine speed is 2,000 rpm	Approximately 3.0V★  PBIB0528E	G

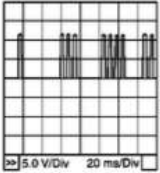
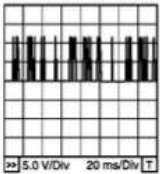
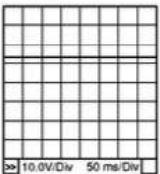
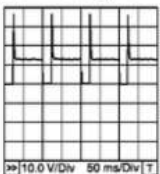
TROUBLE DIAGNOSIS

[CR (WITH EURO-OBD)]

TERMI- NAL NO.	WIRE COLOR	ITEM	CONDITION	DATA (DC Voltage)
22 23 41 42	OR L R GY	Injector No. 3 Injector No. 1 Injector No. 4 Injector No. 2	[Engine is running] ● Warm-up condition ● Idle speed NOTE: The pulse cycle changes depending on rpm at idle.	BATTERY VOLTAGE (11 - 14V)★  PBIB0529E
			[Engine is running] ● Warm-up condition ● Engine speed is 2,000 rpm	BATTERY VOLTAGE (11 - 14V)★  PBIB0530E
24	Y	Heated oxygen sensor 1 heater	[Engine is running] ● Warm-up condition ● Engine speed is below 3,600 rpm.	Approximately 7.0V★  PBIB0519E
			[Engine is running] ● Engine speed is above 3,600 rpm.	BATTERY VOLTAGE (11 - 14V)
29	B	Sensor ground (Camshaft position sensor)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V
30	B	Sensor ground (Crankshaft position sensor)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V
34	OR	Intake air temperature sensor	[Engine is running]	Approximately 0 - 4.8V Output voltage varies with intake air temperature.
35	BR	Heated oxygen sensor 1	[Engine is running] ● Warm-up condition ● Engine speed is 2,000 rpm.	0 - Approximately 1.0V (Periodically change)
45	L	Sensor power supply	[Ignition switch ON]	Approximately 5V
46	W	Sensor power supply (Refrigerant pressure sensor)	[Ignition switch ON]	Approximately 5V
47	L	Sensor power supply (Throttle position sensor)	[Ignition switch ON]	Approximately 5V

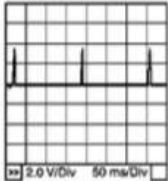
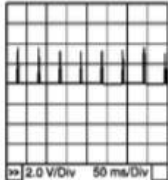
TROUBLE DIAGNOSIS

[CR (WITH EURO-OBD)]

TERMI- NAL NO.	WIRE COLOR	ITEM	CONDITION	DATA (DC Voltage)
14	R	Camshaft position sensor (PHASE)	[Engine is running] ● Warm-up condition ● Idle speed NOTE: The pulse cycle changes depending on rpm at idle.	1.0 - 4.0V★  PBIB0525E
			[Engine is running] ● Engine speed is 2,000 rpm.	1.0 - 4.0V★  PBIB0526E
15	W	Knock sensor	[Engine is running] ● Idle speed	Approximately 2.5V
16	LG	Heated oxygen sensor 2	[Engine is running] ● Warm-up condition ● Engine speed is below 3,600 rpm (A/T models), 3,800 rpm (M/T models) after the following conditions are met. ● Keeping the engine speed between 3,500 and 4,000 rpm for 1 minute and at idle for 1 minute under no load.	0 - Approximately 1.0V
19	LG	EVAP canister purge volume control solenoid valve	[Engine is running] ● Idle speed	BATTERY VOLTAGE (11 - 14V)★  PBIB0050E
			[Engine is running] ● Engine speed is about 2,000 rpm (More than 100 seconds after starting engine)	Approximately 10V★  PBIB0520E

TROUBLE DIAGNOSIS

[CR (WITH EURO-OBD)]

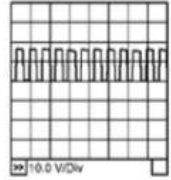
TERMI- NAL NO.	WIRE COLOR	ITEM	CONDITION	DATA (DC Voltage)
49	Y	Throttle position sensor 1	[Ignition switch ON] ● Engine stopped ● Shift lever position: D (A/T models) ● Shift lever position: 1st (M/T models) ● Accelerator pedal: Fully released	More than 0.36V
			[Ignition switch ON] ● Engine stopped ● Shift lever position: D (A/T models) ● Shift lever position: 1st (M/T models) ● Accelerator pedal: Fully depressed	Less than 4.75V
51	W	Manifold absolute pressure sensor	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 1.5V
			[Engine is running] ● Warm-up condition ● Engine speed is 2,000 rpm.	Approximately 1.2V
54	—	Sensor ground (Knock sensor)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V
56	B	Sensor ground (Manifold absolute pressure sensor)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V
57	Y	Sensor ground (Refrigerant pressure sensor)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V
60 61 79 80	Y PU G BR	Ignition signal No. 3 Ignition signal No. 1 Ignition signal No. 4 Ignition signal No. 2	[Engine is running] ● Warm-up condition ● Idle speed NOTE: The pulse cycle changes depending on rpm at idle.	0 - 0.1V★ 
			[Engine is running] ● Warm-up condition ● Engine speed is 2,000 rpm.	0 - 0.2V★ 

PBIB0521E

PBIB0522E

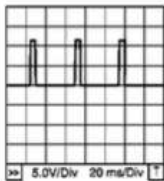
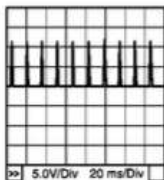
TROUBLE DIAGNOSIS

[CR (WITH EURO-OBD)]

TERMI- NAL NO.	WIRE COLOR	ITEM	CONDITION	DATA (DC Voltage)	
62	LG	Intake valve timing control solenoid valve	[Engine is running] ● Warm-up condition ● Idle speed	BATTERY VOLTAGE (11 - 14V)★	A
			[Engine is running] ● Warm-up condition ● When revving engine up to 2,000 rpm quickly	Approximately 4V - BATTERY VOLTAGE (11 - 14V)★ 	EC C D E
66	B	Sensor ground (Throttle position sensor)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V	F
68	R	Throttle position sensor 2	[Ignition switch ON] ● Engine stopped ● Shift lever position: D (A/T models) ● Shift lever position: 1st (M/T models) ● Accelerator pedal: Fully released	Less than 4.75V	G
			[Ignition switch ON] ● Engine stopped ● Shift lever position: D (A/T models) ● Shift lever position: 1st (M/T models) ● Accelerator pedal: Fully depressed	More than 0.36V	H
69	BR	Refrigerant pressure sensor	[Engine is running] ● Warm-up condition ● Both A/C switch and blower switch are ON (Compressor operates.)	1.0 - 4.0V	I
72	P	Engine coolant tempera- ture sensor	[Engine is running]	Approximately 0 - 4.8V Output voltage varies with engine coolant temperature.	J
73	B	Sensor ground (Engine coolant temper- ature sensor)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V	K
74	B	Sensor ground (Heated oxygen sensor)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V	L
82	B	Sensor ground (APP sensor 1)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V	M
83	B	Sensor ground (APP sensor 2)	[Engine is running] ● Warm-up condition ● Idle speed	Approximately 0V	
85	LG	DATA link connector	[Ignition switch ON] ● CONSULT-II or GST is disconnected.	BATTERY VOLTAGE (11 - 14V)	
86	W	CAN communication line	[Ignition switch ON]	1.0 - 2.5V	
90	OR	Sensor power supply (APP sensor 1)	[Ignition switch ON]	Approximately 5V	

TROUBLE DIAGNOSIS

[CR (WITH EURO-OBD)]

TERMI- NAL NO.	WIRE COLOR	ITEM	CONDITION	DATA (DC Voltage)
91	BR	Sensor power supply (APP sensor 2)	[Ignition switch ON]	Approximately 5V
92	GY	Throttle position sensor signal output (A/T models)	[Ignition switch ON] ● Engine stopped ● Shift lever position is D (A/T models) ● Accelerator pedal: Fully released	Approximately 0.5V
			[Ignition switch ON] ● Engine stopped ● Shift lever position: D (A/T models) ● Accelerator pedal: Fully depressed	Approximately 4.2V
94	R	CAN communication line	[Ignition switch ON]	2.5 - 4.0V
98	GY	Accelerator pedal position sensor 2	[Ignition switch ON] ● Engine stopped ● Accelerator pedal: Fully released	0.3 - 0.6V
			[Ignition switch ON] ● Engine stopped ● Accelerator pedal: Fully depressed	1.95 - 2.4V
101	W	Stop lamp switch	[Ignition switch ON] ● Brake pedal: Fully released	Approximately 0V
			[Ignition switch ON] ● Brake pedal: Depressed	BATTERY VOLTAGE (11 - 14V)
102	GY	PNP switch	[Ignition switch ON] ● Gear position: P or N (A/T models), Neutral (M/T models)	Approximately 0V
			[Ignition switch ON] ● Except the above gear position	BATTERY VOLTAGE (11 - 14V)
103	L/OR	Tachometer signal output (A/T models)	[Engine is running] ● Warm-up condition ● Idle speed NOTE: The pulse cycle changes depending on rpm at idle.	10 - 11V★  MBIB0053E
			[Engine is running] ● Engine speed is 2,000 rpm	10 - 11V★  MBIB0054E
104	G	Throttle control motor relay	[Ignition switch OFF]	BATTERY VOLTAGE (11 - 14V)
			[Ignition switch ON]	0 - 1.0V

TROUBLE DIAGNOSIS

[CR (WITH EURO-OBD)]

TERMI- NAL NO.	WIRE COLOR	ITEM	CONDITION	DATA (DC Voltage)
106	W	Accelerator pedal position sensor 1	[Ignition switch ON] ● Engine stopped ● Accelerator pedal: Fully released	0.6 - 0.9V
			[Ignition switch ON] ● Engine stopped ● Accelerator pedal: Fully depressed	3.9 - 4.7V
109	PU	Ignition switch	[Ignition switch OFF]	0V
			[Ignition switch ON]	BATTERY VOLTAGE (11 - 14V)
111	P	ECM relay (Self shut-off)	[Engine is running] [Ignition switch OFF] ● For a few seconds after turning ignition switch OFF	0 - 1.0V
			[Ignition switch OFF] ● More than a few seconds passed after turning ignition switch OFF	BATTERY VOLTAGE (11 - 14V)
113	R	Fuel pump relay	[Ignition switch ON] ● For 1 second after turning ignition switch ON [Engine is running]	0 - 1.0V
			[Ignition switch ON] ● More than 1 second after turning ignition switch ON	BATTERY VOLTAGE (11 - 14V)
115 116	B B	ECM ground	[Engine is running] ● Idle speed	Engine ground
119 120	G G	Power supply for ECM	[Ignition switch ON]	BATTERY VOLTAGE (11 - 14V)
121	BR	Power supply for ECM (Back-up)	[Ignition switch OFF]	BATTERY VOLTAGE (11 - 14V)

★: Average voltage for pulse signal (Actual pulse signal can be confirmed by oscilloscope.)

CONSULT-II Function (ENGINE) FUNCTION

EBS0003P

Diagnostic test mode	Function
Work support	This mode enables a technician to adjust some devices faster and more accurately by following the indications on the CONSULT-II unit.
Self-diagnostic results	Self-diagnostic results such as 1st trip DTC, DTCs and 1st trip freeze frame data or freeze frame data can be read and erased quickly.*
Data monitor	Input/Output data in the ECM can be read.
Data monitor (SPEC)	Input/Output of the specification for Basic fuel schedule, A/F feedback control value and the other data monitor items can be read.
CAN diagnostic support monitor	The results of transmit/receive diagnosis of CAN communication can be read.
Active test	Diagnostic Test Mode in which CONSULT-II drives some actuators apart from the ECMs and also shifts some parameters in a specified range.
Function test	This mode is used to inform customers when their vehicle condition requires periodic maintenance.
DTC & SRT confirmation	The status of system monitoring tests and the self-diagnosis status/result can be confirmed.
ECM part number	ECM part number can be read.

*: The following emission-related diagnostic information is cleared when the ECM memory is erased.

- Diagnostic trouble codes